Examination of Course: Mathematical Statistics

November 5, 2026

Question 1

1.1 Generate n samples following the distribution $\mathcal{N}(\mu, \sigma^2)$, and calculate the unbiased estimators and confident intervals of μ and σ^2 (both of them are unkown). The true values of n, μ and σ^2 are 100, 1, 5^2 respectively. The significant level is 0.05.

1.2 Repeat the process in **1.1** for 1000 times. We can get 1000 estimated values for μ and σ^2 respectively. Refer to the following definition,

Definition 1. For an estimator $\hat{\theta}$ of θ, its mean squared error (MSE) is defined as

$$MSE_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2].$$

Theorem 1. Specifically, if $\hat{\theta}$ is unbiased then

$$MSE_\theta(\hat{\theta}) = Var_\theta(\hat{\theta}). \tag{1}$$

(1) According to **Definition 1.**, calculate the MSE for both estimators of μ and σ^2.

(2) By further calculating $Var_\theta(\hat{\theta})$, check whether your calculated results can support Equation 1 or not.

Definition 2. An estimator $\hat{\theta} = \hat{\theta}(X)$ is said to be consistent for parameter θ if it converges to θ in probability as $n \to \infty$.

(3) Change value of n in Question **1.1** from 10 to 10000 with a step of 20 and repeat **1.1**, check whether or not estimated values become closer to μ, σ^2 as n increases with the help of **figures**.

1.3 Theorem 2. If $\hat{\theta}$ is a maximum likelihood estimator of θ, then $\hat{\theta}$ is asymptotically normal with mean θ and variance $1/n\mathcal{I}(\theta)$. More precisely, there is convergence in distribution as $n \to \infty$,

$$\sqrt{n\mathcal{I}(\theta)}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, 1),$$

were $\mathcal{I}(\theta)$ is the Fisher information of θ.

Task: Explore this Theorem for the estimator of μ.

Hint: you may plot the distribution of variable and see whether it is similar to a **Standard Normal distribution** or not. The value of n is set to be 10, 100, 1000, and 10000 in order. The sampling process is required to be repeated by 1000 times.

Question 2

In best linear estimation, we learn the best linear estimators for the shift family and scale family. Answer the following questions.

2.1 Suppose that $\boldsymbol{X} = (X_1, \ldots, X_n)$ is a random sample with common density $f_\theta(x)$ ($\theta \in \mathbb{R}$) obtained from a parent density $f_0(x)$ by *shifting*,

$$f_\theta(x) = f_0(x - \theta).$$

with a random variable Z having density $f_0(x)$ and θ as a shift parameter. Simulate data with $f_0(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, $\theta = 3$ and $n = 10$.

(1) Derive the density $f_\theta(x)$.

(2) Calculate the best linear estimator of θ as $\hat{\theta}$ and its MSE using R with simulation. Check whether $\hat{\theta} = \bar{X}$ and $\mathsf{MSE}_\theta(\hat{\theta}) = \sigma^2/n$ hold or not, where $\sigma^2 = \mathrm{Var}(Z)$.

2.2 Suppose that $\boldsymbol{X} = (X_1, \ldots, X_n)$ is a random sample with common density $f_\theta(x)$ ($\theta > 0$) obtained from a parent density $f_1(x)$ by *scaling*,

$$f_\theta(x) = \theta^{-1} f_1(\theta^{-1} x).$$

with a random variable Z having density $f_1(x)$ and θ as a scaling parameter. Simulate data with $f_1(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-1)^2}{2}}$, $\theta = 3$ and $n = 10$.

(1) Derive the density $f_\theta(x)$.

(2) Calculate the best linear estimator of θ as $\hat{\theta}$ and its MSE using R with simulation. Check whether $\hat{\theta} = \frac{\mu}{n\mu^2+\sigma^2} \sum_{i=1}^n X_i = \frac{n\mu}{n\mu^2+\sigma^2} \bar{X}$ and $\mathsf{MSE}_\theta(\hat{\theta}) = \frac{\theta^2\sigma^2}{n\mu^2+\sigma^2}$ hold or not, where $\sigma^2 = \mathrm{Var}(Z)$.

Question 3

In the Chapter Sufficiency, we get to know the Rao–Blackwell theorem.

Rao–Blackwell theorem

Let $\hat{\theta}$ be an unbiased estimator of θ, and let $T = T(X)$ be a sufficient statistic for θ. Consider the conditional expectation

$$\hat{\theta}^* = E_\theta(\hat{\theta}|T).$$

Then $\hat{\theta}^*$ is also an unbiased estimator of θ which is a function of T, and moreover,

$$\mathrm{MSE}_\theta(\hat{\theta}^*) \leq \mathrm{MSE}_\theta(\hat{\theta}).$$

In fact, the inequality is strict unless $\hat{\theta}$ is a function of T.

Answer questions:

Let $\mathbf{X}$ be a random sample from a normal distribution $\mathcal{N}(2\theta, 1)$ (i.e., with unknown mean $2\theta \in \mathbb{R}$ and known variance $\sigma^2 = 1$). The estimator $\hat{\theta} = X_1/2$ is unbiased, since $\mathsf{E}_\theta(X_1) = 2\theta$. The minimal sufficient statistic here is $T_n = X_1 + \cdots + X_n \sim \mathcal{N}(2n\theta, n)$.

(1) Derive the Rao–Blackwell estimator of θ.

(2) Calculate the MSE of the Rao–Blackwell estimator with simulation and compare it with that of the $X_1/2$.

Question 4

Let $X_1, \ldots, X_n$ be i.i.d. from $N(\mu, \sigma^2)$ with μ known. Now, consider the parameter $\eta = 2\sigma$. Derive the Cramér–Rao lower bound for η.

Hint: : consider estimating the variance $\theta = \sigma^2$. The Fisher information is $I_n(\sigma^2) = n/(2\sigma^4)$.

Question 5

Under the Bayesian framework, the unknown parameter θ is treated as a random variable, rather than an unknown deterministic constant. We start with a prior distribution of θ, with density $q(\theta)$ (which can be based on previous information or subjective beliefs), and the focus of estimation is shifted towards updating the prior distribution using the data $X = (X_1, \cdots, X_n)$, leading to the posterior distribution, with density $q(\theta|X)$.

$$q(\theta|X) = \frac{q(\theta)q(X|\theta)}{\int q(\theta)q(X|\theta)d\theta} \propto q(\theta)q(X|\theta),$$

where $q(X|\theta)$ is also the likelihood $L(\theta|X)$.

Consider an i.i.d. random sample $X = (X_1, \cdots, X_n)$ from the Bernoulli distribution Ber(θ) ($0 \leq \theta \leq 1$).

Task 1: Assume that the prior distribution is Beta distribution with parameters 3 and 7. Afterwards, you get some observations from the following process.

(1) Generate n samples from Bernoulli distribution $Ber(0.3)$, where n is 20.

(2) After that, try to get the posterior distribution of θ and plot the distribution.

(3) Explore how posterior expectation and credible interval (credible level 0.9) of θ change as n increases from 20 to 1000 with a step of 20.

Task 2:

(4) Derive the Jeffreys prior in the Bernoulli model for parameter θ, repeat the steps (1)-(3).

(5) Repeat the steps (1)-(3) with flat prior for parameter θ.

(6) Derive the Bayesian mean square errors for the estimated parameter θ with flat prior, and check if the estimators are Bayesian efficient or not, that is, whether its MSE attains the Bayesian lower bound.

Question 6

Part A

Let $X = (X_1, \cdots, X_n)$ be an i.i.d. sample set from the distribution $\mathcal{N}(4, 1)$ with label $y = 1$, and the other be that from distribution $\mathcal{N}(-4, 1)$ with label $y = -1$. If classify sample from the first distribution as -1, we make a mistake and vice versa.

One student decides which group a sample x is generated from by the following rule: if the density $f(x|\mu = 4) > f(x|\mu = -4)$, then classify the sample to group 1, or else to group -1.

(1) Generate 10000 samples from $\mathcal{N}(4, 1)$ and 10000 samples from $\mathcal{N}(-4, 1)$. Calculate the misclassification rate numerically with R and show that the numerical result is similar to the theoretical result.

Hint: the experiment should be conducted several times to make the result more reliable.

(2) Student A generates 100 samples from $\mathcal{N}(4, 1)$ and gives them to student B, A asks B to guess whether those samples are generated from group 1 or not. The two simple hypotheses are

$$H_0 : \mu = 4 \,(\text{null}) \text{ versus } H_1 : \mu \neq 4 \,(\text{alternative});$$

Under the significant level $\alpha = 0.05$, $z_{\alpha/2} = 1.96$, calculate the Type I error numerically with R and show that the result is similar to the theoretical result.

Part B

Let $X = (X_1, \cdots, X_n)$ be an i.i.d. sample set from the distribution $\mathcal{N}(\theta, 1)$, consider simple hypotheses

$$H_0 \colon \theta = \theta_0, \qquad H_1 \colon \theta = \theta_1.$$

Explore the consistency of the Neyman-Pearson Test by both derivation and simulation. **Hint:** : in the simulation, let n=4, 16, 64, 256, 1024 separately, and significant level $\alpha = 0.05$ ($z_\alpha = 1.65$). $\theta_1 > \theta_0$.

Part C

In Bayesian Hypotheses testing, let $\boldsymbol{X} = (X_1, \ldots, X_n)$ be a random sample from

a normal distribution $\mathcal{N}(2\theta, \sigma^2)$, with unknown mean $2\theta \in \mathbb{R}$ and known variance $\sigma^2 > 0$.

Task: Test the hypothesis $H_0 \colon \theta \leq \theta_0$ against $H_1 \colon \theta = \theta_1$.

Hint: Prior distribution can be chosen as $\mathcal{N}(\theta_0, b^2)$. Suppose $\bar{X} = 4$, $\theta_0 = 2.8$.

Question 7

Task. Using the knowledge of mathematical statistics learned in this course, predict the final assignment submission times for all N enrolled students.

Hint: The hard deadline for the final assignment is

$$T_0 = 2026.\ 11.\ 12.\ 17{:}25.$$

N denotes the total number of students enrolled in this course.

Let T_i denote the actual submission timestamp of the i-th student to submit the final assignment, where $i = 1, \ldots, N$ is indexed according to the real-world submission order from earliest to latest.

Define the **relative submission time** (unit: hours) by

$$X_i = \frac{T_0 - T_i}{1 \text{ hour}}, \qquad i = 1, \ldots, N.$$

- If $X_i > 0$, the i-th student submits before the deadline.
- If $X_i = 0$, the i-th student submits exactly at the deadline.
- If $X_i < 0$, the i-th student submits after the deadline.

Since students are indexed according to their actual submission order, the relative submission times satisfy

$$\boxed{X_1 \geq X_2 \geq \cdots \geq X_N.}$$

Here, X_1 is the relative submission time of the earliest student to submit, while X_N is that of the latest student to submit.

Example. For example, a prediction of

$$\hat{X}_i = 0.42$$

means that the predicted submission of the i-th student to submit occurs approximately 0.42 hours before the deadline, whereas

$$\hat{X}_i = -0.12$$

means that the predicted submission occurs approximately 0.12 hours after the deadline.

Requirements.

1. Provide the complete set of N predicted relative submission-time values

$$\hat{X}_1, \hat{X}_2, \ldots, \hat{X}_N.$$

Here, $\hat{X}_i$ denotes the predicted relative submission time of the i-th student to submit.

The predicted values must be ordered from the earliest predicted submission to the latest predicted submission:

$$\boxed{\hat{X}_1 \geq \hat{X}_2 \geq \cdots \geq \hat{X}_N.}$$

2. Explain your statistical reasoning, modelling assumptions, and prediction method in detail. Your grade depends partly on the quality of your statistical argument, and partly on how well your predicted ordered sequence matches the real observed submission pattern after the deadline.

3. Explicit timestamps are not required and will not be used for grading. Your final predictions should be reported as numerical relative submission-time values, for example

$$101.54, \quad 50.98, \quad 20.21, \quad \ldots, \quad -0.12.$$

Evaluation Criterion. The accuracy of your predicted submission-time sequence will be evaluated using the **Wasserstein-1 distance** between the predicted and actual relative submission-time distributions.

Let

$$\hat{X}_1 \geq \hat{X}_2 \geq \cdots \geq \hat{X}_N$$

denote your predicted relative submission times, and let

$$X_1 \geq X_2 \geq \cdots \geq X_N$$

denote the actual relative submission times, both ordered from the earliest to the latest submission.

In this setting, the Wasserstein-1 distance is

$$\boxed{W_1(\hat{X}, X) = \frac{1}{N} \sum_{i=1}^{N} \left| \hat{X}_i - X_i \right|.}$$

A smaller value of W_1 indicates that the predicted submission-time sequence is closer to the actual submission pattern.

Part of the historical data for the year 2025: 114.02, 94.02, 72.68, 51.18, 50.85, 46.47, 46.00, 45.73, 44.00, 43.95, 42.80, 40.95, ... , 20.78, 20.18, ..., 18.38, 17.77, 17.05, 16.78, ..., 3.32, 3.28, ..., 1.47, 1.43, 1.42, 1.20, 1.10, ..., 0.35, 0.33, 0.28, 0.25, 0.22, 0.10, 0.10, 0.07, 0.02, -0.07, -0.07, -2.98). (Submission window: from 2025.10.31 14:00 to 2025.11.14 17:25.)